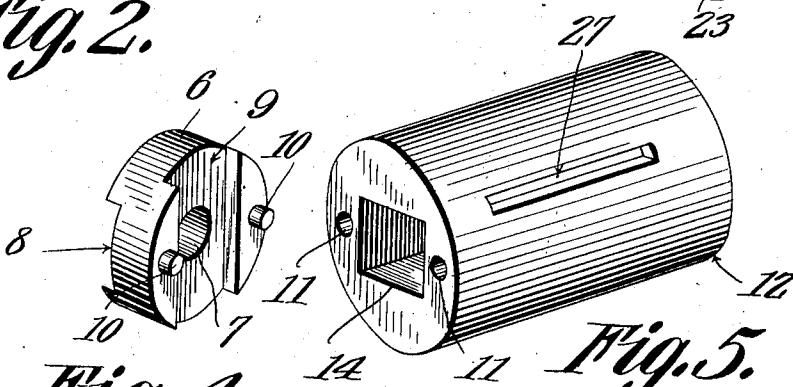
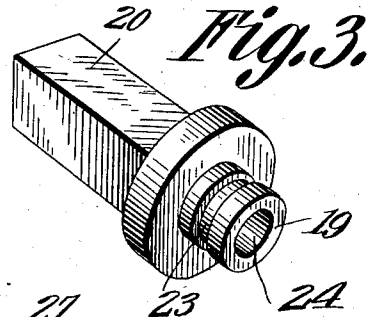
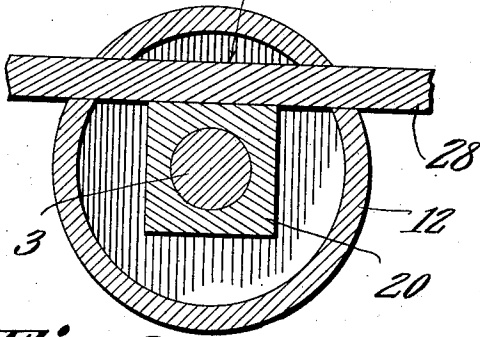
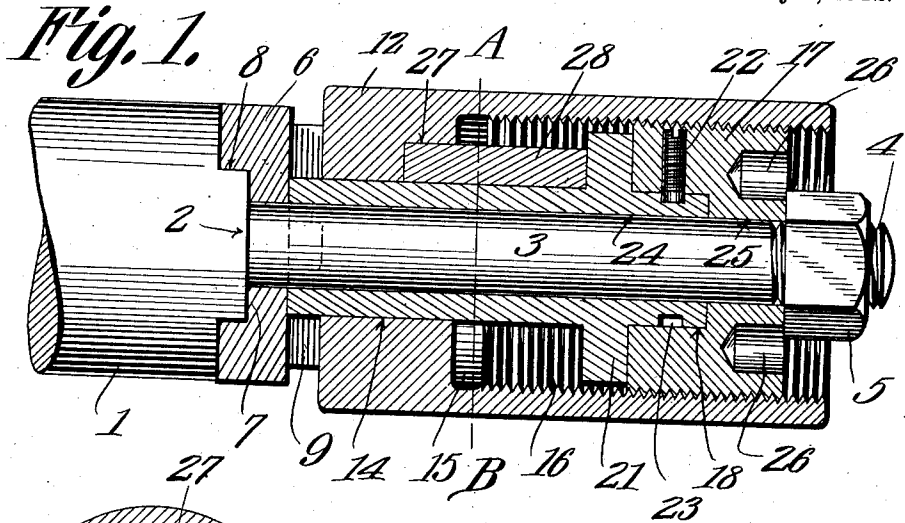


W. F. & J. D. MATTERN.
CUTTER HEAD.
APPLICATION FILED OCT. 4, 1911.

1,025,149.

Patented May 7, 1912.



Witnesses

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UNITED STATES PATENT OFFICE.

WILLIAM F. MATTERN AND JOEL D. MATTERN, OF SHAMOKIN, PENNSYLVANIA.

CUTTER-HEAD.

1,025,149.

Specification of Letters Patent.

Patented May 7, 1912.

Application filed October 4, 1911. Serial No. 652,868.

To all whom it may concern:

Be it known that we, WILLIAM F. MATTERN and JOEL D. MATTERN, citizens of the United States, residing at Shamokin, in the county of Northumberland, State of Pennsylvania, have invented a new and useful Cutter-Head, of which the following is a specification.

The device forming the subject matter of this application, is a cutter head, adapted to hold a tool, for use in jointing, gaining, tenoning, in the fashioning of moldings, and elsewhere, the device being of wide application.

The invention aims to provide a novel means for assembling the head with the spindle of a machine, and to provide novel means for retaining the tool in the head.

In the accompanying drawings,—Figure 1 is a longitudinal section, parts appearing in elevation; Fig. 2 is a transverse section upon the line A—B of Fig. 1; Fig. 3 is a perspective of the stock portion of the follower; Fig. 4 is a perspective of the connection which is interposed between the spindle of the machine and the socket portion of the device; and Fig. 5 is a perspective of the socket.

In the drawings, the numeral 1 denotes the spindle of the machine with which the device is to be assembled, the spindle 1 being provided in its end face, with an outstanding shoulder 2, having an outstanding arm 3, threaded as shown at 4, to receive a nut 5. A connection 6 is provided, having a central opening 7 to receive the arm 3, the connection 6 having in one end, a slot 8, adapted to receive the shoulder 2 of the spindle 1, there being in the other end face of the connection 6, a slot 9, disposed at right angles to the slot 8. Upon either side of the slot 9, pins 10 outstand. The pins 10 are adapted to register in openings 11, formed in one end of a tubular socket 12, having in its end wall, a polygonal opening 14, communicating with a circular chamber 15, internally threaded as shown at 16.

The invention further includes a follower, adapted to move within the tubular socket 12. This follower comprises a circular head 17, threaded to engage with the threaded portion 16 of the socket 12. The inner end of the head 17 is recessed, as shown at 18, to receive for rotation, the circular end 19 of a polygonal stock 20, conforming closely to the opening 14 in the end of the socket

12, the dimensions of the socket 20 being such that it will fit closely, and against rotation, within the slot 9 of the connection 6. The stock 20 is equipped with an intermediate, outstanding flange 21, adapted to lie against the inner end face of the head 17, the circular end 19 of the stock, and the head 17 being connected for independent rotation, by means of a screw 22, which, being threaded radially into the head 17, enters a circumscribing groove 23 in the rounded portion 19 of the stock 20.

Through the entire length of the stock 20 extends a circular opening 24, adapted to receive the arm 3 of the spindle 1, there being in the head 17, a circular opening 25, through which the arm 3 extends, the construction being such that the nut 5 is located adjacent the outer end face of the head 17. In the outer end face of the head 17 there are spaced openings 26, adapted to receive the lugs of a wrench or key, (not shown) whereby the head 17 may be rotated in the socket 12. There is a transverse slot 27 extending through the socket 12, to one side of the center of the socket, and through this slot 27 may be inserted a tool 28, the squared stock 20 lying along one side of the tool 28, as clearly shown in Fig. 2. Since the tool 28 may be of any desired form, the tool has been indicated conventionally at 28, without showing any specific form of cutting edge, since the cutting edge will be varied, in accordance with the work in hand.

The operation of the device is as follows. The nut 5 is loosened, and the lugs of a wrench or key are inserted in the openings 26, causing a rotation of the head 17, whereupon the tool 28 may be inserted in the slot 27. The head 17 may now be rotated, until the flange 21 bears against one edge of the tool, thereby binding the tool firmly in place. When the head 17 is rotated, the squared stock 20 will move longitudinally, but without rotation, in the opening 14 of the socket 12. The squared stock 20 lies along one side of the tool 28, and therefore serves to reinforce the tool in its position within the socket 12. When the head 17 is rotated, in order to cause the flange 21 to bind upon the tool, the inner end of the squared stock 20 will protrude through the opening 14 and register in the transverse slot 9 of the connection 6. Thus, the squared stock 20 will cooperate with the pins 10 in uniting the

socket 12 for simultaneous rotation with the connection 6 and with the spindle 1.

Having thus described the invention, what is claimed is:—

- 5 1. A device of the class described comprising a socket provided with a transverse, tool-receiving slot; and a follower consisting of a head threaded into the socket, and a stock connected with the head for independent rotation, the stock having a flat
10 face, alining with one edge of the slot, and being provided with an outstanding portion adapted to bind against the edge of the tool which is in the slot.
- 15 2. A device of the class described comprising a socket having a tool-receiving slot; a head threaded into the socket; and a stock adapted to engage one edge of the tool which is mounted in the slot, and to extend
20 along one face of the tool; the rotation of the head serving to bind the stock against the tool, and to advance one end of the stock beyond the socket, to afford a connection between the socket and a rotating element, and
25 means for connecting the head and the stock for independent rotation.
3. A device of the class described comprising a socket having a transverse slot for the reception of a tool; a head threaded into
30 the interior of the socket; a stock connected

with the head for independent rotation, and adapted to bear against the tool which is within the slot; the stock having a polygonal portion, adapted to extend beyond one end of the socket, to engage with a rotating
35 element; there being an axial opening through the stock and through the head, to receive a driving spindle.

4. A device of the class described comprising a connection having a transverse
40 slot in one end; a socket having a tool-receiving opening; a head threaded into the socket; a stock connected with the head for independent rotation, and adapted to bind
45 against the tool which is within the opening, the stock having a polygonal portion adapted to register against rotation in the slot of the connection; there being communicating openings through the connection,
50 through the stock, and through the head, adapted to receive the spindle of a driving tool.

In testimony that we claim the foregoing as our own, we have hereto affixed our signatures in the presence of two witnesses.

WILLIAM F. MATTERN.
JOEL D. MATTERN.

Witnesses:

FRANK B. OCHSENREITER,
LENA WILLSON.

Copies of this patent may be obtained for five cents each, by addressing the "Commissioner of Patents, Washington, D. C."